

Active Fiber Tips With Optoelectronic Integration: State-of-the-Art, Future Trends, and Challenges

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(Invited paper)

Abstract—The integration of conventional silica optical fibers with optoelectronic materials and electrical structures opens the door to miniaturized and multifunctional all-fiber optoelectronic systems with actively controllable capabilities. The flat endface of an optical fiber tip offers a naturally light-coupling microscopic platform for optoelectronic integration using optimized micro- and nano-fabrication techniques. In recent decades, the increasing demand for active fiber devices has created opportunities to develop various optoelectronic structures on fiber tips, from integrating high-performance single modules for specific applications to integrating multi-function multiple modules for more complex applications. In this paper, we review the rapid development of fiber-compatible fabrication technologies for building optoelectrical circuits on fiber endfaces. Then, we discuss the fast-progressing field of active fiber tips along with its wide applications, including photodetection, modulation, lasing, sensing, and light-emitting. Finally, we discuss challenges and opportunities for future development in this field. We believe that this scheme of active fiber tips will pave the way for next-generation ultracompact optical fiber systems with diverse optical and optoelectronic applications.

Index Terms—Active devices, optical fiber devices, optical fiber tips, optoelectronic integration.

I. INTRODUCTION

THE invention of low-loss silica optical fibers, semiconductor lasers, and erbium-doped fiber amplifiers laid the foundations of modern highly developed optical fiber communication networks. The unique properties of optical fibers, including their small size, high flexibility and biocompatibility, high mechanical robustness, and electromagnetic immunity, result in a broad range of applications in both fundamental science and everyday life. Over decades, various passive and active optical fiber devices have been developed for various applications. Passive fiber devices use all-optical structures without external sources,

including fiber couplers, inline isolators and circulators, optical attenuators, optical sensors, and polarizers. Although a range of passive fiber components is available to achieve fiber capabilities, much of the signal generation, modulation, processing, and detection is still carried out electrically. Fortunately, active fiber devices have also been well developed; such devices use external sources to excite functional materials for the modulation or monitoring of light propagated in fibers, including fiber lasers, amplifiers, modulators, photodetectors, photoelectrical sensors, and photoswitches. Generally, conventional optical fiber devices tend to be bulky and even require coupling between optical fibers and space optics or on-chip optics, resulting in a complex, high-loss optical fiber system with numerous optical fiber devices used in series or in parallel.

Therefore, the wide demand for miniaturized and multifunctional active devices has promoted the emergence of a new technology for integrated fiber circuits, wherein researchers have proposed two approaches to overcome the fundamental limitations of the material characteristics and geometry structures of silica optical fibers. The first approach is called “multi-materials fibers,” wherein the silica fiber structures have evolved by merging multi-structures and multi-materials (such as metals and semiconductors wires) during the fiber drawing, to produce fibers that can simultaneously see, hear, sense, and communicate [1], [2], [3], [4], [5], [6], [7], [8], [9], [10], [11], [12]. This emerging trend of multi-material fibers has aroused significant industrial interest; however, it also faces some technical challenges, such as mode-field mismatch and fiber splicing difficulties. An alternative approach based on lab-on-fiber (LOF) technology has aroused thriving research momentum, focusing on the modification and processing of commercial silica optical fibers, followed by metal electrode deposition and functional material assembly, to extend their photonic and optoelectronic applications [13], [14], [15], [16], [17], [18].

According to the light-matter interaction approach, LOF-based active fiber devices can be categorized into two groups: guided waveguide integration and fiber endface integration. Waveguide integration refers to the integration of electrode structures and functional materials on the outer surface of optical fiber structures (e.g., microfibers [19], [20], unclad fibers, and D-shaped fibers [21], [22], [23], [24]) or within the fiber structures (e.g., micro-structured fibers [25], [26]), with efficient light coupling over long distances and large areas. However, the cylindrical structure of the optical fiber strongly limits the

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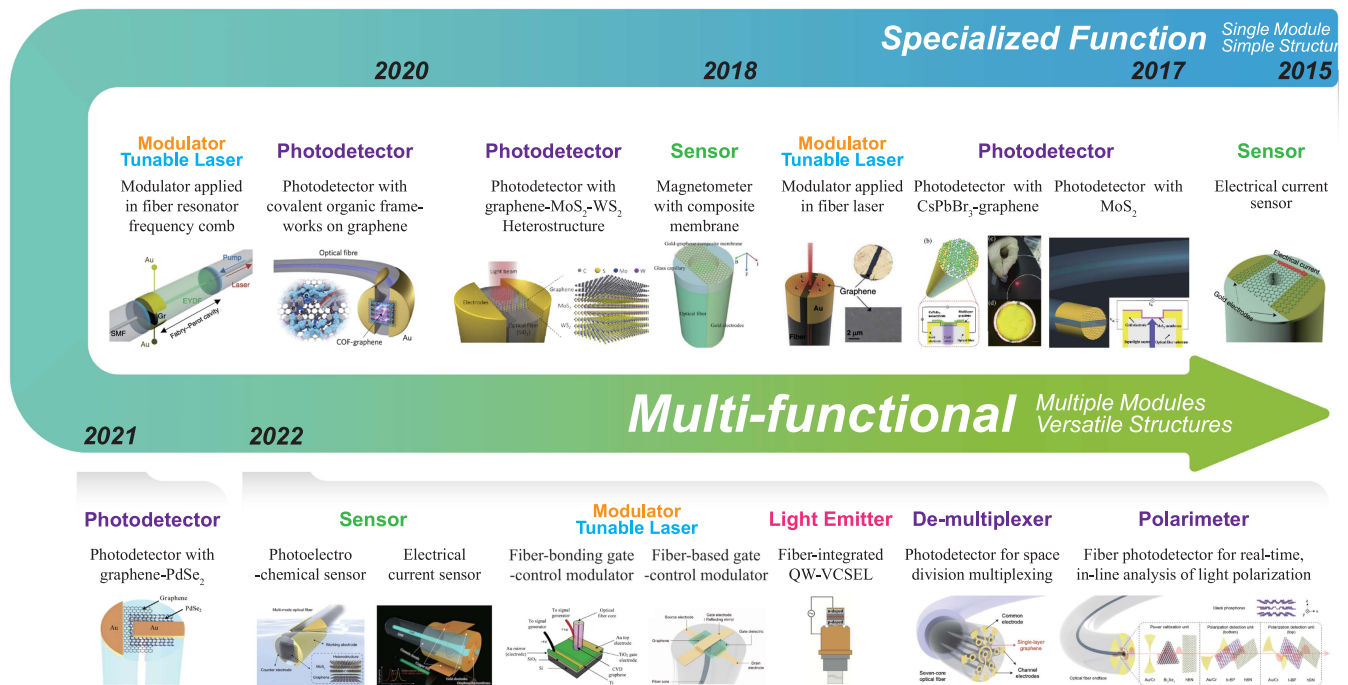


Fig. 1. Summary of the state-of-the-art active fiber tips.

compatibility of traditional planar micro- and nano-fabrication technologies with the fiber sidewall for waveguide integration. In contrast, the microscale plane of the fiber endface has emerged as a new integration platform, showing potential for applications in micro- and nanotechnologies. Fiber endface devices possess nearly free-space light coupling along with self-aligned optical paths, serving as integrated plug-and-play components for optical networks. However, when we treat the fiber endface as a planar substrate, its small diameter with a large aspect ratio still presents challenges when adapting standard nano-fabrication strategies.

In this paper, we first review the two main integration approaches reported thus far for active fiber tips, namely the direct processing of electrode structures and materials on fiber tips and the assembly of pre-prepared planar chips onto fiber tips. Subsequently, we present a detailed overview of the state-of-the-art active fiber tips and their wide applications (Fig. 1). Finally, we propose the ongoing development of strategies aiming to build multifunctional active fiber tips.

II. INTEGRATION TECHNOLOGIES

The unique shape of the optical fiber, with a small diameter and large aspect ratio, drives the optimization of standard planar nanostructure fabrication strategies for the optoelectronic integration on fiber tips, which can be categorized into two main technical approaches. The first approach focuses on preparing the electrode structure and transferring functional materials directly on the fiber tips via optimized optical fiber-compatible techniques. This approach is compact and reliable, with the flexibility of optical fibers preserved to the maximum extent; however, it usually suffers from the availability of fabrication

instruments designed specifically for planar substrates, low production efficiency, and limited repeatability owing to the lack of standard fiber fixtures for mass production. In the second approach, the electrode structure and functional materials are first fabricated on a planar chip using standard planar nanotechnologies, showing potential for mass production. The main challenge is the subsequent assembling of the prepared chip onto the fiber tip while ensuring that the completed structure is firmly bonded to the fiber endface and aligned precisely with the fiber core.

A. Direct Processing on Fiber Tips

In recent decades, several LOF techniques have been proposed for the direct processing of micro- and nanostructures on fiber tips, including typical techniques such as mechanical processing, focused ion beam (FIB) etching, lithographic techniques, and transfer methods. As for active fiber tips, the key fabrication procedures include the patterning of designed electrode structures and precise transfer of functional optoelectronic materials.

1) *Patterning of Electrodes:* As shown in Fig. 2, several techniques have been proposed thus far for patterning electrodes on fiber tips, from creating simple dual electrodes to building multiple modules with stacked structure and customized geometry, accompany with the evolution of fabrication techniques.

In earlier studies, dual electrode structures on fiber tips were prepared via simple mechanical processing [27], [28], [29], [30], [31]. Chen et al. proposed a typical procedure [27], wherein an optical fiber tip was first coated with a metal film (usually a gold film), with no adhesion layer (usually Cr or Ti) deposited in the middle, to facilitate the subsequent removal of the metal layer. Afterwards, two parallel lapping films were used to remove a

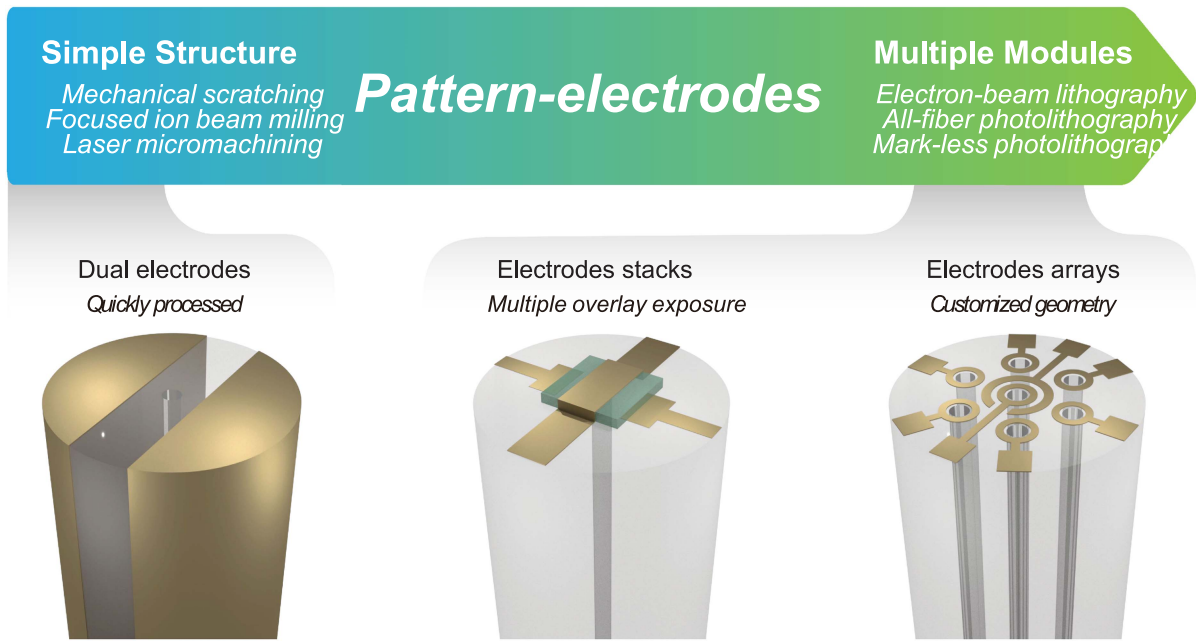


Fig. 2. Summary of the fabrication processes for patterning electrodes on fiber tips.

portion of the metal film on the fiber sidewall via polishing, and a hard metal probe (such as tapered tungsten needles) was used to remove the metal film on the fiber endface via scratching. As a result, a pair of metal electrodes was prepared on the fiber tips with an electrode gap at both the fiber sidewall and endface. It is a quick and easy processing method; however, it is difficult to obtain fine electrode structures and control fabrication consistency and repeatability. To address this problem, FIB milling and laser micromachining have been used to remove the metal film on the fiber endface with improved fabrication accuracy [32], [33], [34]. However, these techniques may damage the smoothness of the fiber substrate and are often time-consuming for high-accuracy processing. Alternatively, physical masks have been used to realize patterned simple electrode structures on the fiber endfaces [35], [36]. For convenience of operation, a ceramic ferrule with a large planar area (typical diameter ~ 2.5 mm) was used to immobilize the fiber instead of using the optical fiber itself (typical diameter ~ 125 μm). The physical mask was aligned with the fiber core and adhered to the endface of the fiber with an outer ceramic ferrule. After subsequent metal deposition process, the mask pattern was transferred to the fiber endface. This method is simple and efficient; however, the precise alignment of the optical fiber and physical mask is challenging. Moreover, due to the gap that may exist between the optical fiber and the physical mask, the edges of the pattern will not be sharp enough to cause structure errors.

With the development of nanotechnology, optical fiber-compatible lithographic techniques have been widely developed. The fiber endface can be regarded as a relatively large planar surface to embrace lithographic technologies, which also face several challenges owing to the unique shape of optical fibers. Obtaining a uniform, controllable resist layer on an optical fiber endface is one of the most challenging procedures. For bare

optical fibers with a diameter of ~ 125 μm , a resist-coating technique called “dip and vibration” has been proposed [37], [38], [39]. The optical fiber was covered with a thick resist film by dip coating it into the resist solution. Then, the optical fiber held by the fiber holder was mechanically vibrated to remove redundant resist on the endface. The thickness of the resist layer can be controlled by adjusting the frequency and amplitude of the mechanical vibrations of the optical fiber. In addition, spin-coating is the most common technique for resist coating, which can precisely control the thickness of the resist layers via the spinning time and rotation rate. However, the small diameter and large aspect ratio of bare optical fibers make it difficult to adapt the traditional spin-coating method to the optical fiber endface. Therefore, optical fibers immobilized in a ceramic ferrule are often used instead of bare optical fibers, which have a large planar area of ~ 2.5 mm. Moreover, special rotating chucks that hold fibers in the vertical direction must be optimized. Xiong et al. customized an optical fiber holder to achieve a flat multi-layer resist film on a fiber endface. The plane of the fiber endface was level with the plane of the metal holder, which can be seen as an extension of the endface of the fiber from ~ 2.5 mm to ~ 1 cm, similar to a plane substrate.

Several lithography techniques have been proposed for metal electrode preparation based on resist-coating techniques. Photolithography is a powerful approach for micro- and nano-fabrication. Petrušis et al. proposed an all-fiber photolithography technique using a commercial optical fiber splicing machine [40]. They used a UV-transmitting fiber with a lithographic mask fabricated on its endface as a mask fiber and another fiber coated with a photoresist layer as the target fiber. These two fibers were aligned and brought into precise contact, and ultraviolet light was coupled from the opposite end of the mask fiber, exposing the target fiber. The pattern on the mask fiber can

be transferred to the target fiber after subsequent development, metallic, and lift-off procedures. In addition, maskless photolithographic systems serve as variable photomasks, patterning the fiber endface flexibly [41]. Limited by the wavelength of light, it is difficult to achieve pattern accuracy below the micron level by photolithography techniques. With the development of nanotechnology, lithographic methods have progressed from optical-based photolithography to technologies that use electron beams as radiation sources, significantly improving the precision of the patterns. In the electron beam lithography (EBL) technique, a beam of electrons is focused and scanned on an electron beam resist with an ultimate resolution of sub-10 nm, chemically modifying the resist layer and ultimately forming the predesigned pattern after the development process. To avoid electrostatic charging caused by the electrical insulation of optical fibers, conductive layers such as thin metal layers or conductive polymer layers must be pre-coated on the fiber endface [37], [42], [43]. This direct writing method provides a precise, flexible, and versatile approach for the design and fabrication of unique electrode structures according to fiber structures and application requirements. Recently, Xiong and co-workers demonstrated a precise multiple overlay lithographic process using EBL and prepared three vertically stacked layers of electrodes on a fiber endface [44]. Moreover, according to the unique size and structure of a multi-core fiber, they prepared a special design for arrayed electrode structures, using the EBL method, including a common electrode and seven channel electrodes [45].

2) *Materials Transfer Techniques*: In addition to the aforementioned methods for fabricating electrode structures on optical fiber endfaces, functional optoelectronic material transfer methods have been extensively developed for building complete active fiber tips. To date, numerous nanomaterials, including nanoparticles, quantum dots, nanowires, nanotubes, and 2D materials have been transferred to optical fiber endfaces through van der Waals interactions. Different material transfer methods have been proposed for different nanomaterial dispersion systems, including the dip-coating [27], [28], [46], [47], wet-transfer [48], dry-transfer [35], [44], and micromanipulation methods [30].

The dip-coating method is suitable for transferring nanomaterials from solutions or suspensions. For example, Chen et al. adopted a simple dip-coating method to transfer floated chemical vapor deposition (CVD)-grown 2D material films such as multi-layer graphene or molybdenum disulfide (MoS_2) to the fiber endface without any support polymers such as poly (methyl methacrylate) (PMMA). To reduce the wrinkles of the transferred 2D material and improve the transfer success rate, Xiong et al. used the standard wet-transfer method to transfer graphene onto the fiber endface with the assistance of the PMMA support polymer layer [45]. The fiber lifted the PMMA-graphene film floating on the water surface from the bottom up, obtaining a uniform wrinkle-free single-layer graphene film on the fiber endface. Although the above-discussed two methods can transfer a large area of nanomaterials onto the fiber endfaces, it is challenging to control the size and shape of the materials.

As a precise approach, the dry-transfer method is particularly suitable for mechanically exfoliated or grown crystal

materials. Xiong et al. reported the use of polydimethylsiloxane (PDMS) to dry transfer CVD-grown bismuth selenide (Bi_2Se_3) nanocrystals, mechanically exfoliated black phosphorus (BP) crystals, and mechanically exfoliated hexagonal boron nitride (hBN) crystals on the electrodes on a fiber endface, forming a six-layer van der Waals heterostructure [44]. Additionally, for fragile layered grown materials, a precise transfer method named the “micro-manipulation method” was reported for building a graphene- MoS_2 - WS_2 heterostructure film onto a fiber endface with controllable size and shape [30].

B. Assembling Planar Chips

Benefiting from the well-developed nano-fabrication techniques available for the planar platform, assembling planar chips on a fiber endface is another potential route for building active fiber tips. Recently, Ricciardi et al. presented the bonding of a chip-integrated quantum-well-based laser on the core of a multimode optical fiber, realizing electrical contact between the chip and the fiber (Fig. 3) [34]. This method requires precise micromanipulation to ensure coaxiality between the fiber cores and chips, which directly influences the laser in-coupling efficiency.

To solve the problem of coupling, Lau et al. fabricated on-chip wafer-scale graphene electro-optic modulators, which can be easily combined with single-mode fibers without the need for free-space optical alignment. They used a hardened epoxy polymer with an open aperture of $125\ \mu\text{m}$ on the modulator as a fiber fixture to fit the optical fiber, ensuring an overlap between the fiber core and the modulator area on the chip [49], [50], [51]. This method sets requirements of manufacturing precision and shows limitations when using more complex device and fiber structures. For instance, Umezawa et al. reported an on-chip 2D photodetector array that can be directly coupled to multi-parallel beams from multi-core and few-mode fibers for compact terminal photoreceivers [52]. However, the above-mentioned method of using fiber fixtures does not work in this situation. The main challenge is the highly efficient coupling between the 2D photodetector array and multiple fiber cores.

In general, active devices prepared by assembling optical fibers and planar chips show great advantages, such as wafer-scale production, mature processing technologies, and easier circuit lead-out methods. However, the alignment of fibers and chips remains challenging, and the fiber device loses its flexibility to some extent.

As a brief summary, the selection of fabrication techniques mostly depends on the requirements of applications. For applications where simple electrode structures are required, mechanical method or using a physical mask is a quick and easy approach. However, for electrode structures that require precise alignment and multiple overlay exposure, EBL and photolithography are more standard, stable, and controllable processes. For some applications that do not require high alignment, especially for few/multi-mode fibers, assembling planar chips is a good technical route.

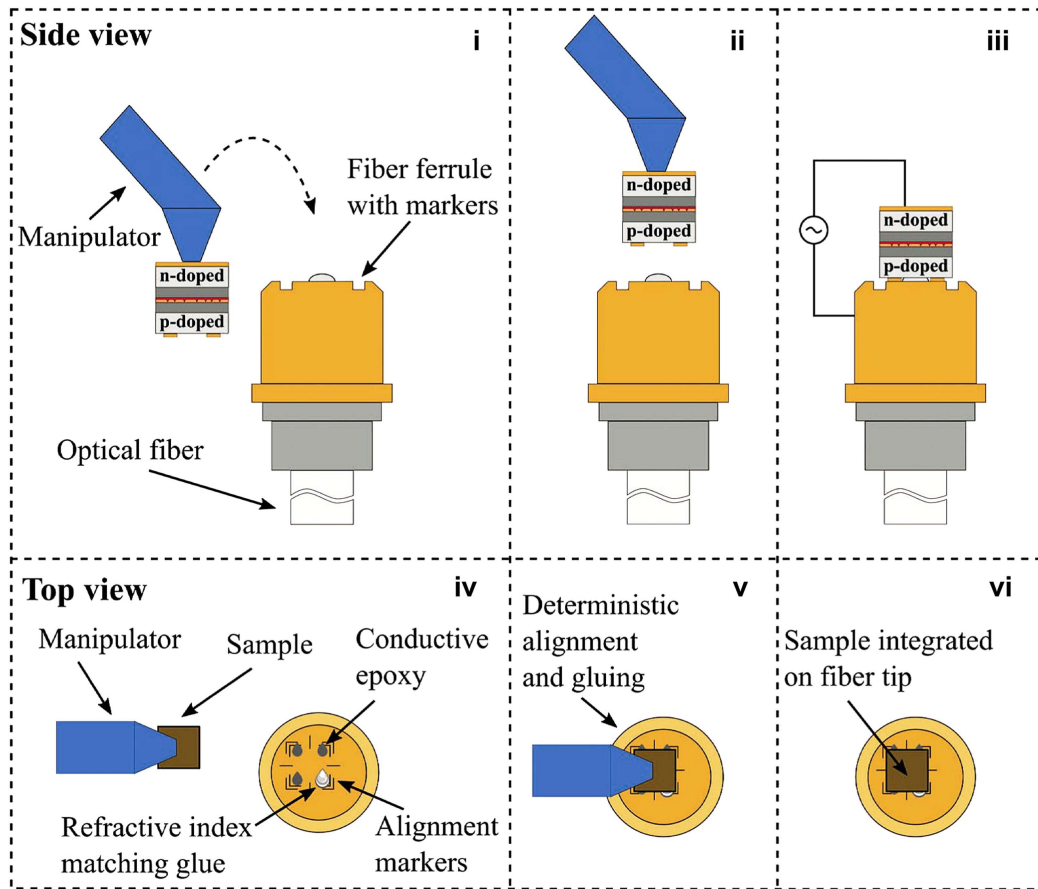


Fig. 3. Schematic illustration of the procedure to integrate planar chips onto a fiber endface from side views (i-iii) and top views (iv-vi) [34]. Copyright 2022, Wiley-VCH GmbH.

III. APPLICATIONS

The previous section introduced numerous studies on the integration of electrical circuits and functional materials onto fiber tips. In this section, we review some typical active device structures on fiber tips and their known potential applications. We first introduce several recent works that integrate single high-performance optoelectronic modules on the fiber endface according to different application requirements, including photodetection, modulation, tunable laser, sensing, and light-emitting. In face of more complex requirements, such as the simultaneous monitoring and regulation of multiple optical signals, it is difficult to achieve using a single module on a fiber tip. Therefore, some emerging works integrate multiple modules onto the fiber endface for multifunctional applications, which we review later.

A. Single Module Integration

1) *Photodetectors*: Photodetectors convert light signals into electrical signals for further processing using standard electronic circuits. Commercial photodetectors are based on bulk semiconductors and detect outcoupled optical signals from optical fibers. The development of novel semiconductor materials, especially low-dimensional materials, offers new possibilities

to realize ultra-compact photodetection units onto the natural light-coupled platform of fiber endfaces, which are called “all-fiber photodetectors (FPD)”. Chen et al. first proposed a visible-light-sensitive FPD by fabricating a MoS_2 photodiode on a fiber endface (Fig. 4(a)) [27]. The total size of the device was almost the same as that of a bare optical fiber and showed a maximum photoresponsivity of $\sim 0.6 \text{ A} \cdot \text{W}^{-1}$ at a bias voltage of 4 V. To further improve the device performance, van der Waals heterostructures were employed as photo-sensing materials. For example, Chen et al. combined graphene and perovskite (CsPbBr_3 nanocrystals) onto a pair of electrodes on a fiber endface to construct a high-performance photodetector (Fig. 4(b)) [28]. Owing to the strong scattering effect of the thick CsPbBr_3 film, nearly all the light power at a wavelength under 520 nm was absorbed, resulting in a low detection limit of 10^{-11} W and photoresponsivity as high as $2 \times 10^4 \text{ A} \cdot \text{W}^{-1}$ at 400 nm illumination.

However, individual light-harvesting semiconductor materials do not possess simultaneously sufficiently high photore-sponsivity, fast response, and broadband working bandwidth. The previous FPDs were only sensitive to visible light because of the limitations of the materials, hindering their application in communication bands. To extend the operating wavelength of a device to the infrared band while maintaining the device performance, Xiong et al. integrated a microscale multi-layer

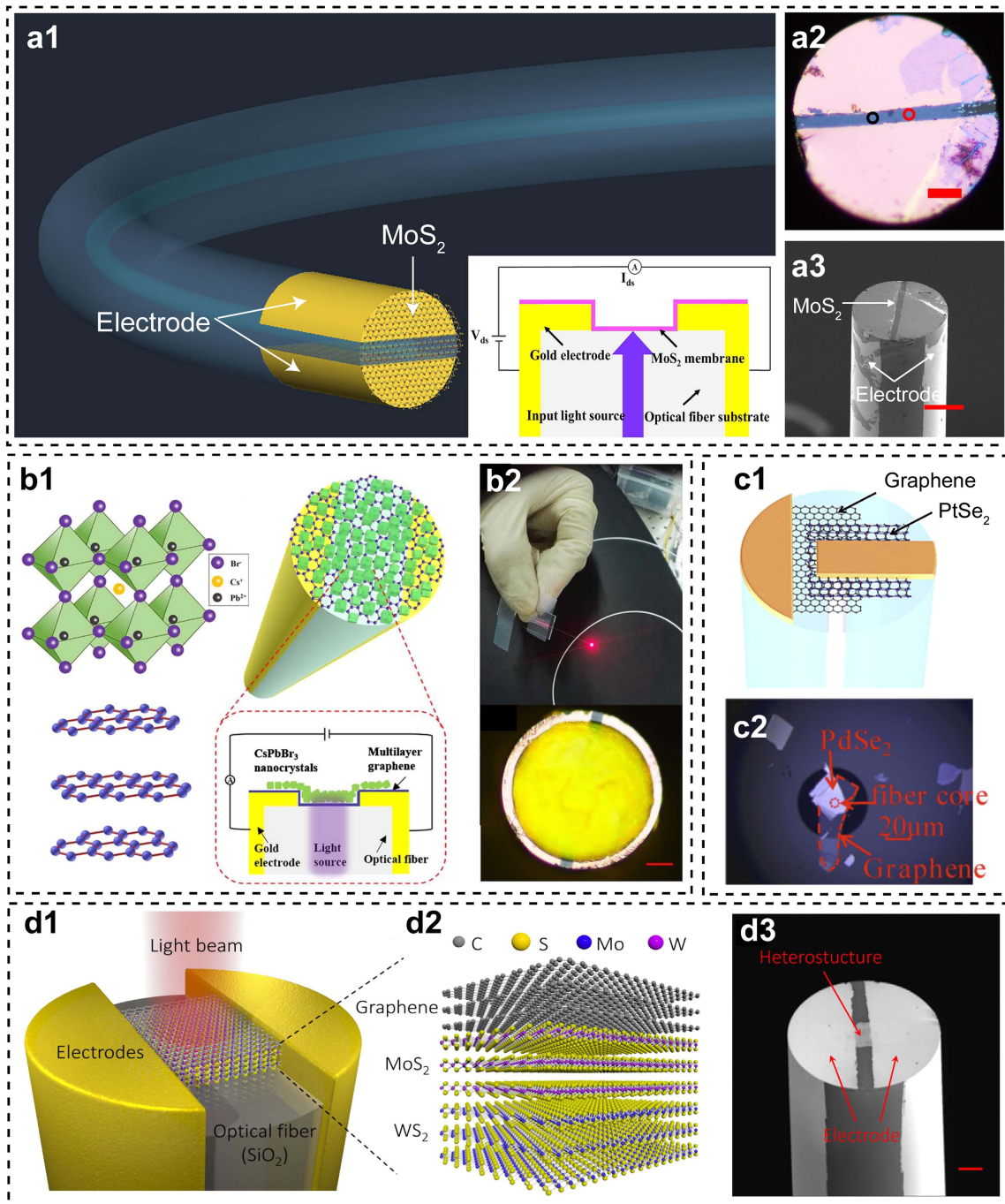


Fig. 4. Fiber-endface-integrated photodetectors. (a) FPD integrated with multi-layer MoS₂ [27]. (a1) Schematic of the FPD. Inset: Cross-sectional part of the FPD. (a2) Optical microscope image of the FPD. (a3) SEM image of the FPD. Reproduced with permission from the Royal Society Of Chemistry. (b) FPD integrated with CsPbBr₃-graphene hybrid structure [28]. (b1) Schematic of the materials and device configuration of the FPD. (b2) Photograph and microscope image of the FPD. Copyright 2017 Optical Society of America. (c) FPD integrated with graphene-PtSe₂ heterostructure [35]. (c1) Schematic view of the device. (c2) Microscope image of the as-fabricated device. Copyright 2021 Optica Publishing Group. (d) FPD integrated with graphene-MoS₂-WSe₂ heterostructure. [30] (d1, d2) Schematic representation of the FPD. (d3) SEM image of the FPD. Copyright 2018 Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim.

graphene-MoS₂-WSe₂ heterostructure film on the fiber endface using a layer-by-layer transfer method (Fig. 4(d)) [30]. Owing to the high carrier mobility inside graphene, strong light absorption of MoS₂-WSe₂, and built-in electric field of the heterostructure, the FPD exhibited an ultrahigh photoresponsivity of $\sim 6.6 \times 10^7 \text{ A} \cdot \text{W}^{-1}$ and a relatively fast time response of $\sim 7 \text{ ms}$ at a wavelength of 400 nm. The interlayer optical transition, due

to the type-II staggered band alignments between the MoS₂ and WSe₂ layers, enabled the FPD to sense 1550 nm infrared light with a photoresponsivity of $\sim 17.1 \text{ A} \cdot \text{W}^{-1}$. Alternatively, as shown in Fig. 4(c), Yang et al. proposed an FPD by assembling a vertical graphene/palladium diselenide (PdSe₂) van der Waals heterostructure onto the endface of an optical fiber [35]. The narrow bandgap of PdSe₂ benefits broadband photodetection

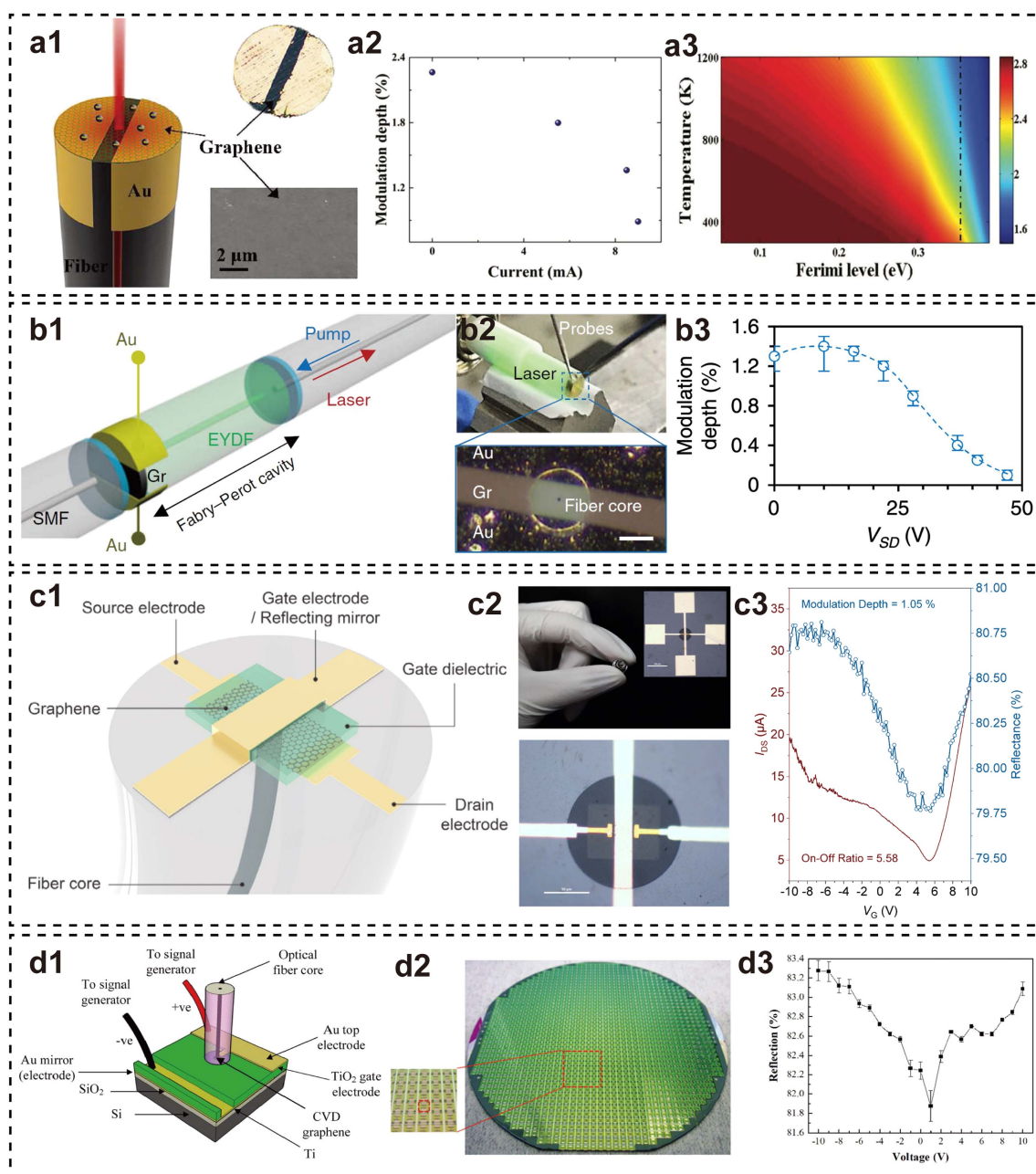


Fig. 5. Fiber-endface-integrated modulators. (a) Fiber-end-face-dual-electrodes thermal modulator based on graphene [29]. (a1) Schematic of the device. (a2) Measured modulation depth of graphene under different driven current. (a3) Theoretical modulation depth of graphene at different fermi levels and temperatures. Copyright 2017 Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim. (b) Fiber-end-face-dual-electrodes graphene modulator incorporated in active fiber FP cavity [36]. (b1) Schematic of the active fiber cavity and configuration of the modulator. (b2) Photograph of cavity (upper) and enlarged microscope image of the modulator (lower). The scale bar in lower panel is 60 μm . (b3) Measured modulation depth of graphene decrease with driven voltage. Copyright 2020, Springer Nature. (c) Fiber-end-face gated-graphene modulator directly fabricated on fiber [64]. (c1) Schematic picture of the device. (c2) Whole-view of the device (upper) and microscope image of the modulator area (lower). The scale bar in lower panel is 50 μm . (c3) Optical reflectance (blue curve) and drain-source current (red curve) tuned by gate voltage. Copyright 2022 Wiley-VCH GmbH. (d) Fiber-end-face gated-graphene modulator fabricated on silicon wafer [50]. (d1) Schematic of the modulator structure. (d2) Mass production of modulators on a 4-inch silicon wafer. (d3) Optical reflection tuned by the electrical voltage applied to the modulator. Copyright 2021 The Royal Society of Chemistry.

from 650 to 1550 nm with a high photoresponsivity of $6.68 \times 10^4 \text{ A}\cdot\text{W}^{-1}$ and a fast temporal response of $\sim 660 \mu\text{s}$. Moreover, the strongly asymmetrical device structure enables the FPD to operate at zero bias, which shows an on/off ratio of ~ 82 . There are still many technical issues to be solved for high-performance photodetection in fiber endface platforms compared with the

planar platform [53], [54] because the insulating silica optical fiber has no back-gate electrodes allowing to modify the material characteristics, and the current fiber platforms lack sophisticated micro-/nano-fabrication technologies. Combining advanced nanostructures and device designs in fiber endface platforms is an intriguing area for further exploration of

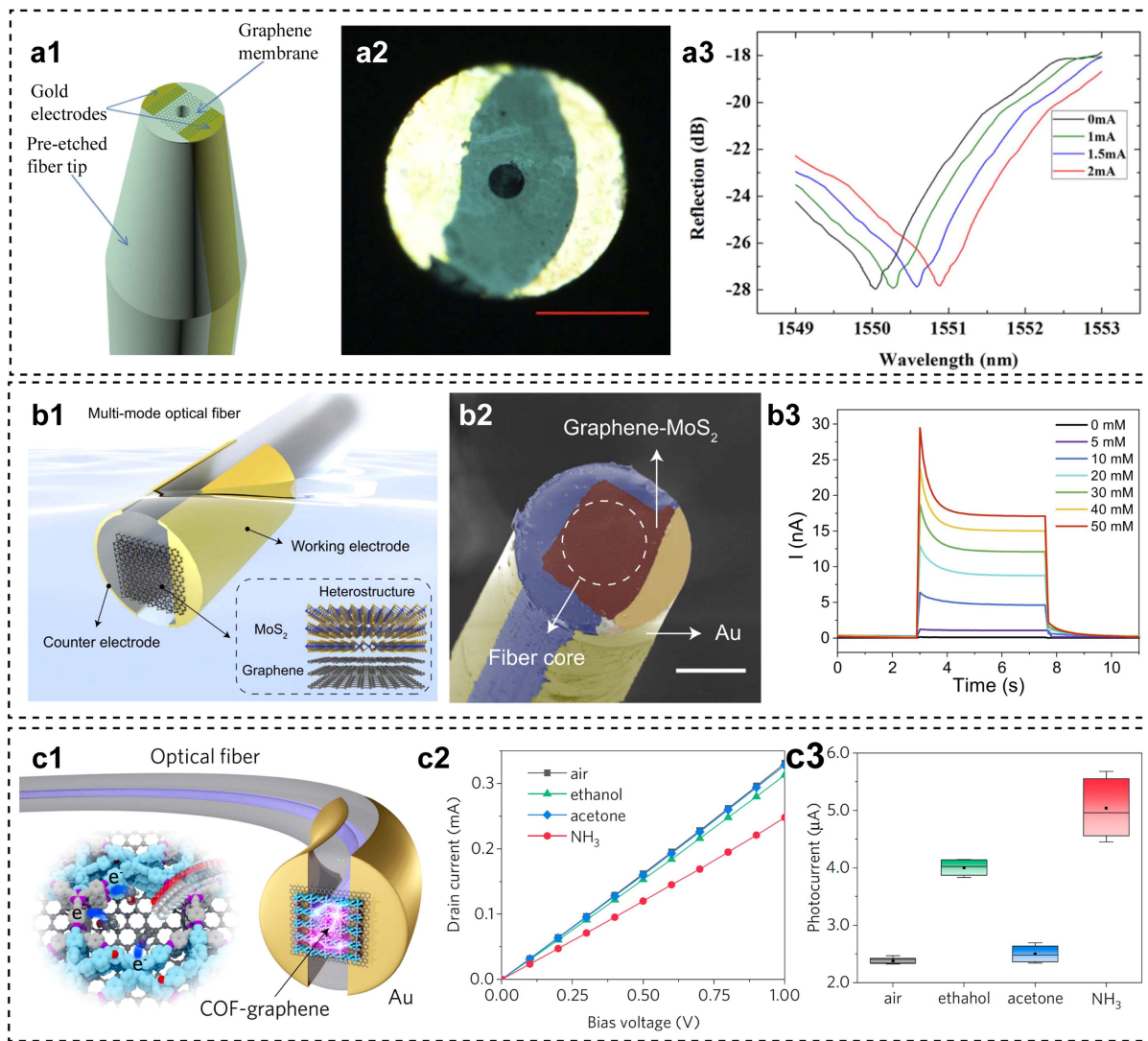


Fig. 6. Fiber-endface-integrated sensors. (a) Miniature optical fiber current sensor based on graphene NOMES [47]. (a1) Schematic of the sensor. (a2) Micrograph of the sensor. (a3) Shifted reflection spectrum with the change of the current. Copyright 2015 Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim. (b) Optical fiber tip integrated photoelectrochemical sensors [32]. (b1) Schematic of the device. (b2) Photocurrent generation mechanism of the device for the detection of AA in solution. (b3) Dynamic current curves in the presence of different amounts of AA. Copyright 2022 Optica Publishing Group. (c) Photoelectrical sensors based on 2D COFs integrated on graphene. (c1) Schematic view of the molecule absorption and charge transfer at the surface of the functional film. (c2) I - V curves with different gas molecules. (c3) Photocurrent generation in different gas atmospheres. Copyright 2020 Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim.

high-performance and multifunctional photodetection [32], [44], [45], [55], [56], [57], [58], [59], [60], [61].

2) *Modulators*: Optical modulators are essential components of photonic and optoelectronic systems. Light propagating through a certain material can be encoded by modifying external factors, such as the electric field, optical field, magnetic field, thermal field, and mechanical strain [62], [63]. The tight optical confinement in the optical fiber core and the plug-and-play property make the optical-fiber endface an ideal platform for fabricating optical modulators. With the development of processes for fabricating electrodes on optical fiber endfaces, it is possible to achieve widely preferred electrically controlled modulators. For example, as shown in Fig. 5(a), Li et al. demonstrated an electro-thermo-optic modulator that actively modulated the nonlinear optical properties of a graphene-fiber endface-integrated

device by modifying the Fermi-Dirac distribution of a graphene sheet using an electric heating method [29]. When the applied electrical current across the graphene sheet was tuned from 0 to 9 mA, the temperature of the graphene changed from ~ 300 to ~ 900 K. Owing to the thermal relaxation process, the nonlinear optical absorption of graphene is relevant to its temperature, showing an electro-thermo-optic modulation depth variation of over 60%. This modulator was then incorporated into a fiber laser circuit to constitute a tunable fiber laser system, which is discussed later. Similarly, Qin et al. demonstrated electrical tuning of a graphene sheet on a fiber endface (Fig. 5(b)) [36]. Once driven by an electrical current, external electrons are injected into the graphene sheet, resulting in a shift in the Fermi level of graphene from p-doping to n-doping. This tunable graphene sheet was employed on the endface of a 1 cm

long erbium/ytterbium co-doped single-mode fiber, constituting a heterogeneous Fabry-Perot resonator for electrically controllable mode-locked laser combs.

Besides the approach using electrical current injection, Ding et al. proposed a fiber-endface-integrated electro-optic modulator based on the electrical gating effect (Fig. 5(c)) [64]. By introducing workflows of silicon-on-insulator processes, a graphene field-effect transistor with a reflecting structure is prepared on a single-mode fiber endface, showing a vivid optical modulation with a modulation depth of over 1% along with the tuning of the gate voltage. Alternatively, researchers have proposed another solution of a gate-controlled device where the modulator itself need not be fabricated directly on the fiber endface. As mentioned above, the devices can be fabricated on a silicon wafer using a mature semiconductor process and then bonded to the fiber tip, which is suitable for mass production. Lau et al., adopting this technical route, demonstrated a scalable graphene electro-optical modulator (Fig. 5(d)) [50]. The graphene-based modulator was designed as a reflective multilayer-gating structure with a dielectric layer placed between a mirror and graphene; the modulation depth reached $\sim 1.4\%$ when the gate voltage was tuned.

3) *Tunable Lasers*: The fiber-end-face-integrated modulator can be easily incorporated into a fiber laser system, thereby enhancing the multifarious tunability of the laser parameters. The interaction types of these modulators can be classified into two modes: passive and active.

In the passive mode, the functional 2D material in the modulator (graphene, e.g.,) acts as a saturable absorber in the laser cavity to enable passive mode locking. By tuning the optical nonlinearity, the passive mode-locking state is changed and regularly controlled by an external signal. In the work of Li et al., the operation regime of the fiber laser was smoothly shifted from mode-locking to continuous wave by controlling the modulation depth of the saturable absorption through the injected current. Qin et al. utilized a similar structure, but applied the device to a short fiber Fabry-Perot laser cavity. The driving current changed the linear and nonlinear absorption of graphene simultaneously, with the former enabling the tuning of the cavity Q-factor and the latter leading to controllable harmonic mode-locking. The central wavelength was tuned by heating the distributed Bragg reflector mirror induced by the current Joule heat.

Modulators in the active mode shows more focus on linear absorption and dynamic responses instead of nonlinear properties. Lau et al. demonstrated the application of a graphene electro-optical modulator in an actively Q-switched $2\ \mu\text{m}$ fiber laser [50]. An electrical square-wave signal was applied to the modulator to achieve periodic intracavity loss switching, thereby realizing the output of a Q-switched pulse train with a repetition rate of 56.5 kHz and a signal-to-noise ratio of 46.6 dB. The relative work also implies the potential of such a modulator to achieve mega-hertz-rate-level active mode-locking pulse generation [49].

4) *Photoelectrical Sensors*: Photoelectrical sensors transform environmental stimuli into a modified light signal or material optoelectrical property which ultimately manifests in

the collected electrical signals. The introduction of electrode structures into silica optical fibers offers different sensing mechanisms, such as sensing based on micro- and nano-opto-electro-mechanical systems (MOEMS/NOEMS), photoelectrochemical (PEC) sensing, and sensitive optoelectronic-material based sensing.

As shown in Fig. 6(a), Zheng et al. integrated a quasistatic graphene-based NOEMS on a fiber endface for electric current sensing [47]. A single-mode optical fiber was etched by an HF-water solution to form a hole in the fiber core area, followed by the fabrication of a pair of gold electrodes on opposite sides of the hole. Subsequently, a suspended multi-layer graphene membrane was transferred onto the endface, and the hole was covered. When an electric current is applied to the graphene membrane, its shape changes owing to the current-induced variation in temperature, which is called the negative thermal expansion effect. The magnitude of the electric current was related to the offset of the reflection spectrum of the sensor. The sensor overcomes the shortcomings of conventional optical fiber current sensors based on thermal effects, showing a high sensitivity of $2.2 \times 10^5\ \text{nm}\cdot\text{A}^{-2}$, short response time of $\sim 0.25\ \text{s}$, and compact device size of $\sim 15\ \mu\text{m}$. Similarly, this device configuration has been further demonstrated for magnetic-field sensing based on the Lorentz force under an electric current. Liu et al. fabricated a fiber-capillary-graphene-gold hybrid structure based on NOEMS, demonstrating a current-driven Lorentz-force magnetometer [65]. The difference is that the hole in the fiber core was replaced by a capillary with a larger hole, and a graphene membrane was used to support a 100 nm thick Au layer, which provided a higher light reflection ratio, lower membrane resistance, and better production repeatability. The shape of the free-standing graphene-gold-composite film changes owing to the Lorentz force induced by the propagating current in the magnetic field. By detecting the shift of the sensor's reflection spectra, the magnitude of the magnetic field was measured with a sensitivity of 0.64 pm/mT and response time of $\sim 0.1\ \text{s}$.

Recently, a compact fiber-integrated PEC sensor was reported [32]. Compared to conventional electrochemical and optical methods, PEC sensing separates the excitation source (light) and detection signal (electric current), showing great potential for low-background signal biochemical molecular detection with high sensitivity. However, traditional PEC sensing systems have large separated structures of electrodes and light sources. To solve this problem, Wang and co-workers integrated a pair of gold electrodes consisting of a working electrode and a counter electrode on a multimode fiber endface (Fig. 6(b)). The working electrode was connected to a transferred graphene-MoS₂ heterostructure membrane and isolated from the counter electrode, where the transparent graphene layer served as an extension of the working electrode, and the MoS₂ layer generated photoexcited electron-hole pairs and interacted with biochemical molecules in solution. Under illumination from the fiber core, an electrical potential difference between the working and counter electrodes is established, which is related to the concentration of target biomolecules and generates different photocurrents in the external circuit. This highly integrated, flexible, and

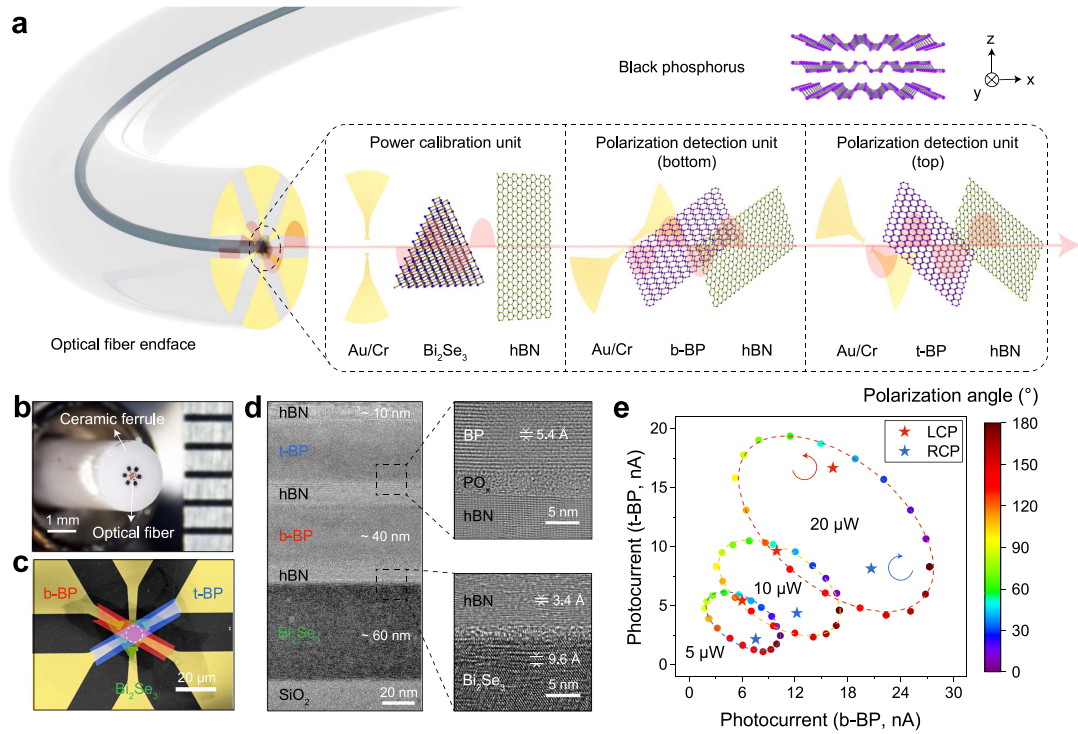


Fig. 7. Serial modules integrated on a fiber endface for light polarization detection [44]. (a) Schematic illustration of the designed device. Three functional modules are stacked subsequently onto an optical fiber endface. (b) Camera image of the polarimeter. (c) Pseudo-color SEM image of the stacked structures on the optical fiber endface. (d) Cross-sectional TEM image of the device. (e) Photocurrent generated in the two twisted BP units under different LP and CP states. Copyright 2022 the authors, published by AAAS.

cost-effective device was suitable for the real-time monitoring of weak biochemical changes in small areas, showing a self-powered detection of ascorbic acid with a concentration range between 1 and 50 mM and a time response of ~ 6 ms. By designing PEC reactions and integrating various recognition elements on the fiber endface, such devices will exhibit selectivity for biomolecules.

Sensitive optoelectronic materials that naturally show response to the environment have been also used to build photoelectrical sensors. As displayed in Fig. 6(c), Xiong et al. designed and fabricated a fiber-integrated photodetector based on graphene and 2D covalent organic frameworks (COFs), which have been proven to be ideal materials for gas adsorption due to their highly porous structure with high surface areas [31]. The photosensing properties of the device can be reversibly regulated by specific target molecules, especially some strongly polar molecules. The gas sensitivities of 1% NH_3 in air and 1% ethanol vapor in air were measured to be 16.2% and 5.8%, respectively.

5) *Light-Emitting Devices*: Generally, optical fibers are still used in combination with external bulk sources with “off-fiber” electrically driven components, which impacts the overall device footprint and achievable scalability, and faces the challenge of fiber-to-chip coupling. Recently, Ricciardi et al. proposed a highly efficient and position-tolerant coupling approach that transferred, aligned, and bonded a quantum-well-based light source onto the core of a multimode optical fiber [34]. The planar chip bonded on the fiber endface showed both electrical contact

between the chip and ferrule (holding the fiber) and a light in-coupling efficiency of $\sim 20\%$. This method can be applied to other optoelectronic chips (such as light-emitting diodes and photodetectors) for specific applications.

B. Multiple-Module Integration

To improve the degree of integration, the compression and combination of traditional functional opto-electro-mechanical components into a single compact device will change the existing paradigm of discrete optical systems, paving the way toward the next generation of multifunctional optoelectronic fiber-integrated devices.

1) *Serial Module Integration*: The fiber end-face naturally provides a light path perpendicular to its plane. Serial integration is more similar to the extreme compression of tandemly arranged bulky space-optics components and integrates them onto the fiber endface.

Recently, Xiong and co-workers designed and fabricated an ultracompact fiber-integrated polarimeter using serial module integration (Fig. 7) [44]. Conventional polarimeters suffer from a series of bulky opto-electro-mechanical elements tandemly arranged along the optical path, which are difficult to minimize and integrate for compact applications [66], [67]. As shown in Fig. 7, three functional photodetection units were successively stacked onto the fiber endface, including a polarization-insensitive isotropic Bi_2Se_3 -based power calibration module

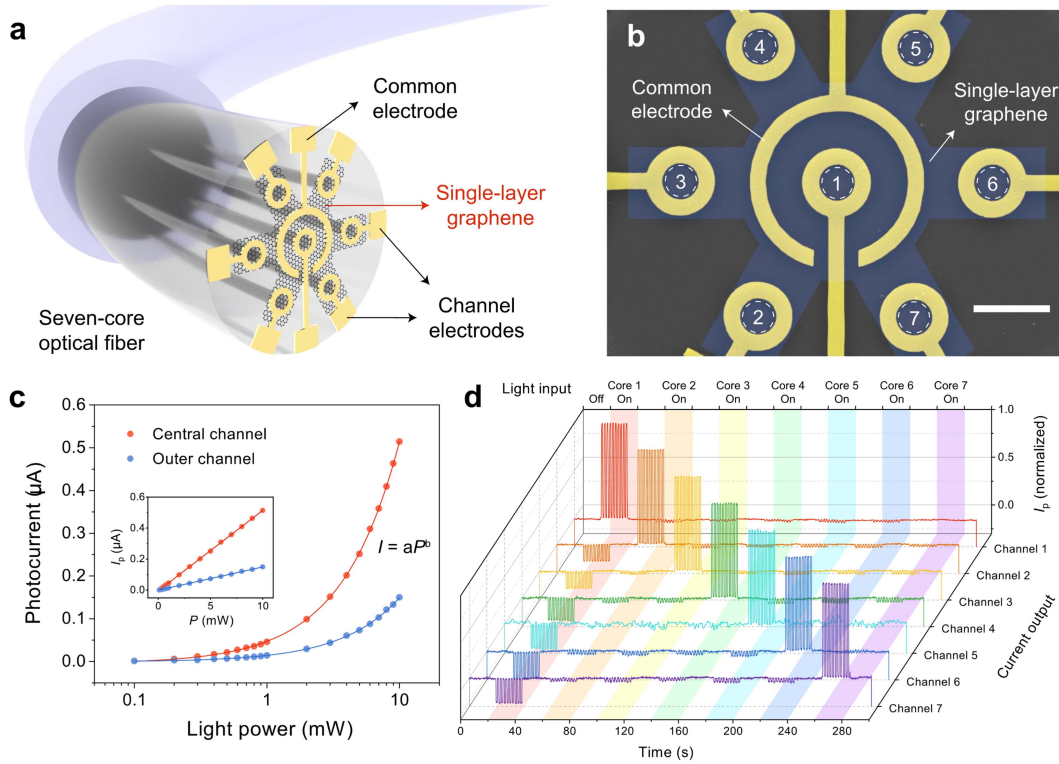


Fig. 8. Parallels modules integrated on a seven-core fiber endface for de-multiplexing [45]. (a) Schematic illustration of the integrated parallel graphene photodetection modules on a commercial seven-core optical fiber. (b) Pseudo-color SEM image of the device. The yellow area presents the Au/Ti electrodes and the blue area presents the patterned graphene film. The scale bar is 20 μm . (c) Spontaneous photocurrents generated in different modules as a function of illuminating power in a semi-log coordinate and a linear coordinate (inset). (d) Seven core de-multiplexing. The figures from top to bottom are the collected photocurrent outputs from channels 1-7, measuring the light response when cores 1-7 are illuminated individually. Copyright 2022 American Chemical Society.

at the bottom and two BP-based polarization detection modules at the top. Each module consisted of a pair of electrodes prepared via the EBL process, a layer of optoelectronic materials for functional photodetection, and a layer of hexagonal boron nitride (hBN) for encapsulation. The electrodes were spatially asymmetrically designed against the fiber core, which broke the system symmetry, enabling self-driven detection of both linearly polarized (LP) and circularly polarized (CP) light. The two BP-based polarization detection modules were twisted and stacked to substitute for the traditional mechanical rotation of the polarizers. By simultaneously obtaining the photocurrent values of these serially arranged modules along the optical paths, the polarization state of an unknown LP or CP light can be unambiguously detected using the device.

Note that the scheme of serially stacked ultrathin modules along the optical path is expected to build multifunctional, highly integrated fiber devices, paving the way for next-generation ultracompact optical systems with diversified optical and optoelectronic applications.

2) *Parallel Module Integration*: For some multiplexing cases, we need to measure the signals of multiple channels simultaneously; parallel module integration provides an effective solution [68]. Multi-core fibers are an essential space-division multiplexing solution for versatile applications [69], [70], [71], [72], [73], [74], [75], [76], [77], [78], [79], [80], [81].

However, coupling and de-multiplexing in a multi-core fiber system remains a challenge [52], [82].

As shown in Fig. 8, Xiong et al. demonstrated an ultra-compact seven-core fiber (SCF) de-multiplexer with parallel module integration, which downsizes traditional bulky modules to the micrometer scale with atomic thickness [45]. The parallel photodetection of seven individual fiber cores was achieved using a patterned single-layer graphene photodetector array prepared on the fiber endface. With precise fabrication through the EBL process, a simplified electrode configuration with a strongly asymmetrical structure was successfully prepared on patterned graphene, thereby realizing self-powered seven-core photodetection. Moreover, the high transparency of single-layer graphene ensures a high transmittance of over 96% in the device, benefiting in-line, real-time light status analysis.

The parallel modules integration can make sufficient usage of the area of the fiber endface, integrating functional modules such as sensing, detection, modulation, and signal processing together.

IV. CONCLUSION

In this decade, active fiber devices have made great progress, from theoretical calculations and device designs to fabrication techniques and measuring methods, with the aim of achieving actively controllable all-fiber systems with compact footprints.

Although we have been able to precisely fabricate versatile electrode structures on optical fibers, many technical challenges remain to be addressed. Unlike silicon-based planar devices, silica optical fibers cannot be used as back-gate electrodes to modify the properties of optoelectronic materials, which induces difficulties in realizing field-effect devices [83]. Due to that conventional film deposition techniques, such as atom layer deposition (ALD) and plasma enhanced chemical vapor deposition (PECVD), are mostly designed for planar substrate. The deposition of uniform conductive and insulating films even 2D materials films on the endface and cylindrical surfaces of optical fibers remains to be explored [84], [85]. Moreover, it is difficult to achieve nano-level flatness on the surface of optical fibers compared to silicon wafers, which also affects device performance owing to the requirement of the atomic-level flat and pure surface for obtaining high-performance. The inherent limitations of a short light-matter interaction distance and small sensing area of the fiber endface integration are a challenge to further improve the device performance, where the integration of nanostructures and nanocavities can solve this problem while the device bandwidth is reduced [86], [87]. Developing batch reproduction processes with reduced costs and increased yields is the main challenge for fiber-based integrated devices, where large-area photolithography and high-quality material growth techniques may be applied [25], [84], [85]. However, the reliable packaging of these devices remains to be addressed for practical applications.

With regard to potential prospects, several trends and directions show great promise:

- 1) the integration of complex electrode structures and auxiliary planar optical and electronic elements on fiber endfaces for high-speed optical modulators, phototransistors, and sensors [88];
- 2) the integration of adjustable optical systems on fiber endfaces for optical field regulation devices, endoscopy probes or light parameters conversion systems [86], [87];
- 3) the integration of unique device structures on specific optical fibers for diversified multiplexing functions [68];
- 4) the integration of functional active modules onto the subwavelength-scale optical fiber tips for cell-level *in vivo* analysis [89].

We believe that the quick development of active fiber tips will open the door to next-generation multifunctional, ultra-compact optical fiber systems.

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